

Scaling laws

Berkan Ottlik

Lecture outline

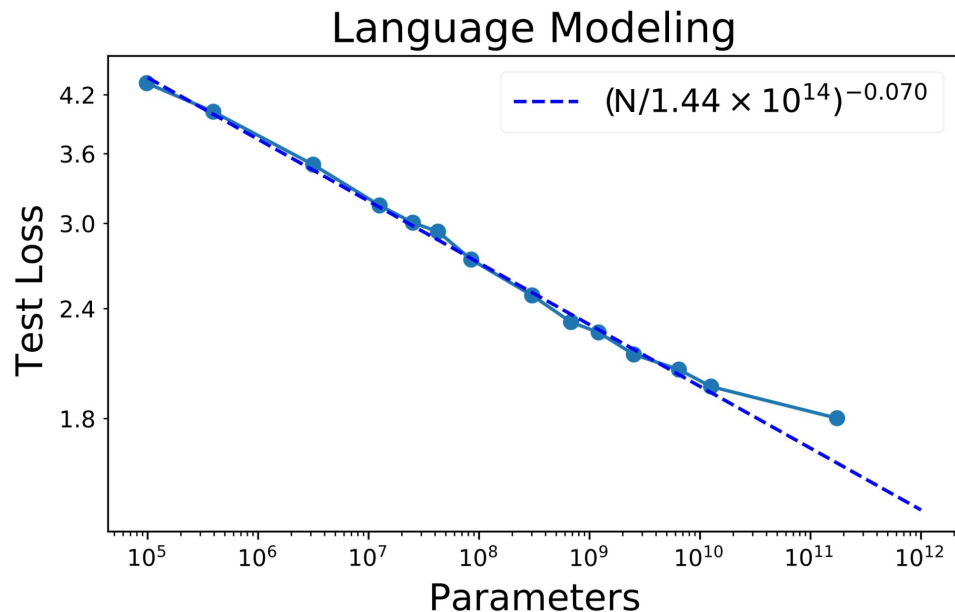
1. Basics, history, and motivation
2. Empirics for pre-training
3. Theory
 - a. Quanta model
 - b. Deriving scaling exponent from data
4. Bonus
 - a. Post-training
 - b. Inference

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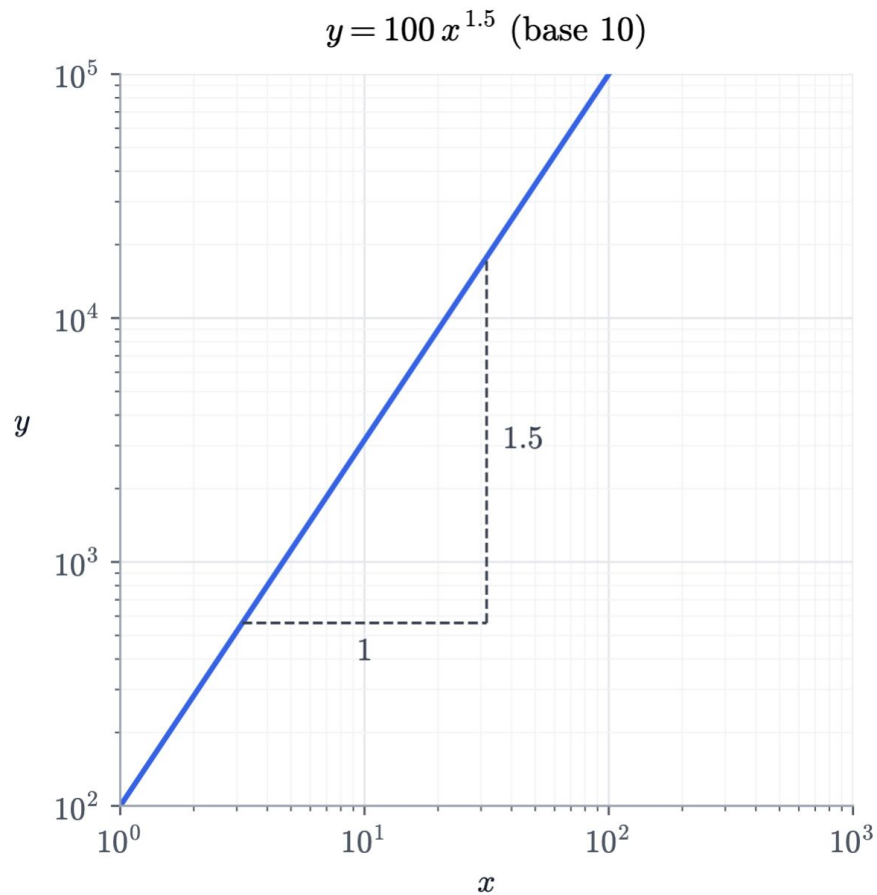
Scaling law

Predicts LLM performance with jointly scaled sources, e.g., data D , compute C , and parameters N .



Power laws

$$y = bx^m$$



Loss curve

Olmo 3 base 7B
pre-training loss
curve.



TV generalization bounds

2024

Theorem 3.1 (Finite-class MLE; TV consequence of [3, Proposition 2.1]). *Let $\mathcal{Q}$ be finite and contain the source distribution p . Let $\hat{q}$ minimize empirical next-token log loss on n independent documents from p . For $\varepsilon, \delta \in (0, 1)$, if*

$$n \geq \frac{2 \log(|\mathcal{Q}|/\delta)}{\varepsilon^2}, \tag{2}$$

then $\text{TV}(p, \hat{q}) \leq \varepsilon$ with probability at least $1 - \delta$ over the training sample.

Learning Curves: Asymptotic Values and Rate of Convergence 1993

Corinna Cortes, L. D. Jackel, Sara A. Solla, Vladimir Vapnik,
and John S. Denker
AT&T Bell Laboratories
Holmdel, NJ 07733

Abstract

Training classifiers on large databases is computationally demanding. It is desirable to develop efficient procedures for a reliable prediction of a classifier's suitability for implementing a given task, so that resources can be assigned to the most promising candidates or freed for exploring new classifier candidates. We propose such a practical and principled predictive method. Practical because it avoids the costly procedure of training poor classifiers on the whole training set, and principled because of its theoretical foundation. The effectiveness of the proposed procedure is demonstrated for both single- and multi-layer networks.

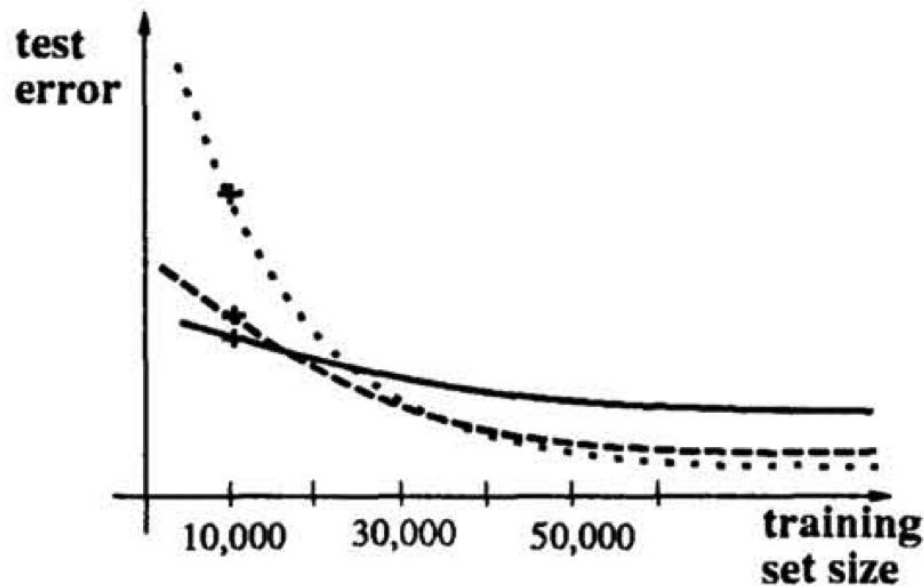


Figure 1: Test errors as a function of the size of the training set for three different classifiers. A classifier choice based on best test error at training set size $l_0 = 10,000$ will result in an inferior classifier choice if the full database contains more than 15,000 patterns.

The Bitter Lesson

Rich Sutton

March 13, 2019

The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin. The ultimate reason for this is Moore's law, or rather its generalization of continued exponentially falling cost per unit of computation. Most AI research has been conducted as if the computation available to the agent were constant (in which case leveraging human knowledge would be one of the only ways to improve performance) but, over a slightly longer time than a typical research project, massively more computation inevitably becomes available. Seeking an improvement that makes a difference in the shorter term, researchers seek to leverage their human knowledge of the domain, but the only thing that matters in the long run is the leveraging of computation.

DEEP LEARNING SCALING IS PREDICTABLE, EMPIRICALLY

2017

**Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun,
Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, Yanqi Zhou**

{joel, sharan, ardalaninewsha, gregdamos, junheewoo, hassankianinejad,
patwarymostofa, yangyang62, zhouyanqi}@baidu.com

Baidu Research

“In particular, accurately predicting generalization error scaling with training set size would provide a powerful tool for estimating the costs—in data and compute requirements—for advancing state-of-the-art (SOTA).”

“Our results show that power-law learning curves exist across all tested domains.”

OpenAI codebase next word prediction

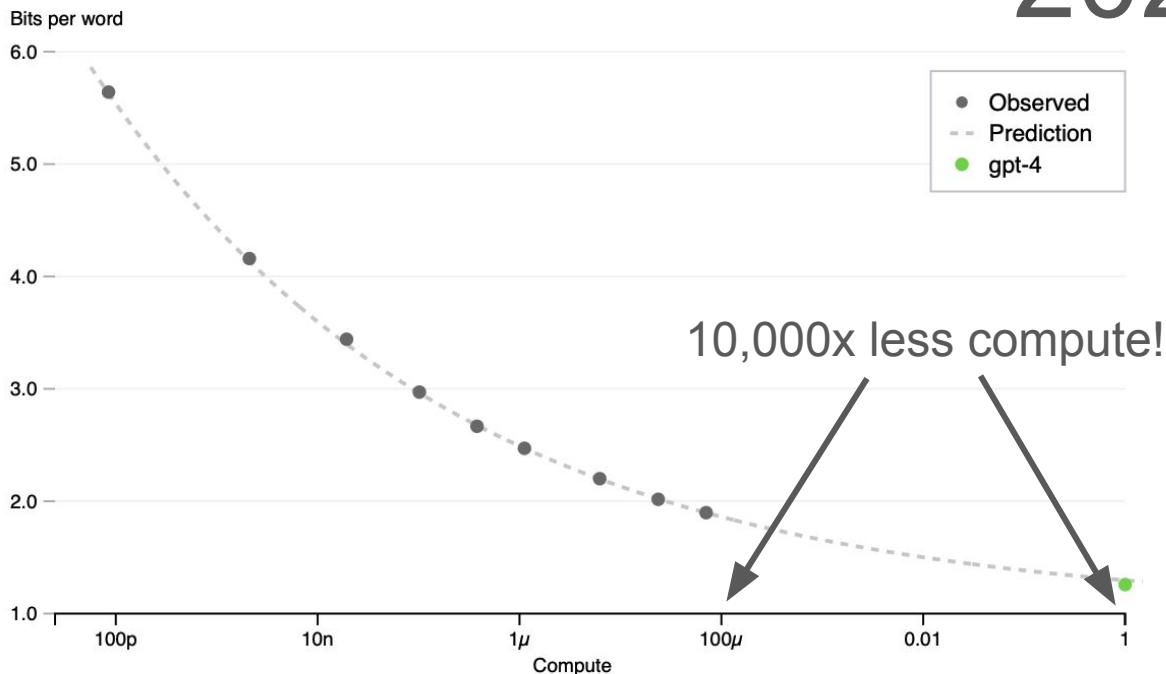


Figure 1. Performance of GPT-4 and smaller models. The metric is final loss on a dataset derived from our internal codebase. This is a convenient, large dataset of code tokens which is not contained in the training set. We chose to look at loss because it tends to be less noisy than other measures across different amounts of training compute. A power law fit to the smaller models (excluding GPT-4) is shown as the dotted line; this fit accurately predicts GPT-4's final loss. The x-axis is training compute normalized so that GPT-4 is 1.

Scaling laws enable cheap large scale
model selection.

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Training Compute-Optimal Large Language Models

Jordan Hoffmann*, Sebastian Borgeaud*, Arthur Mensch*, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals and Laurent Sifre*

*Equal contributions

Compute-optimal pre-training

3 resources: data D , compute C , and parameters N

With a limited compute budget what's the best model you could pre-train?

Compute-optimal pre-training

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With a limited compute budget what's the best model you could pre-train?

$$C = \text{FLOPs}(D, N) \approx 6DN$$

Deriving loss scaling parametric form

Symbol	Meaning
C	Training compute in FLOPs, with $C \approx 6DN$
D	Number of training tokens (data)
N	Number of model parameters
L	Expected loss
$\hat{L}$	Predicted expected loss
f^*	Bayes classifier
f_N	Best transformer with N parameters
$\hat{f}_{N,D}$	Empirical risk minimizing transformer with N parameters, trained on D tokens

$$\hat{L}(N, D) = ?$$

Deriving loss scaling parametric form

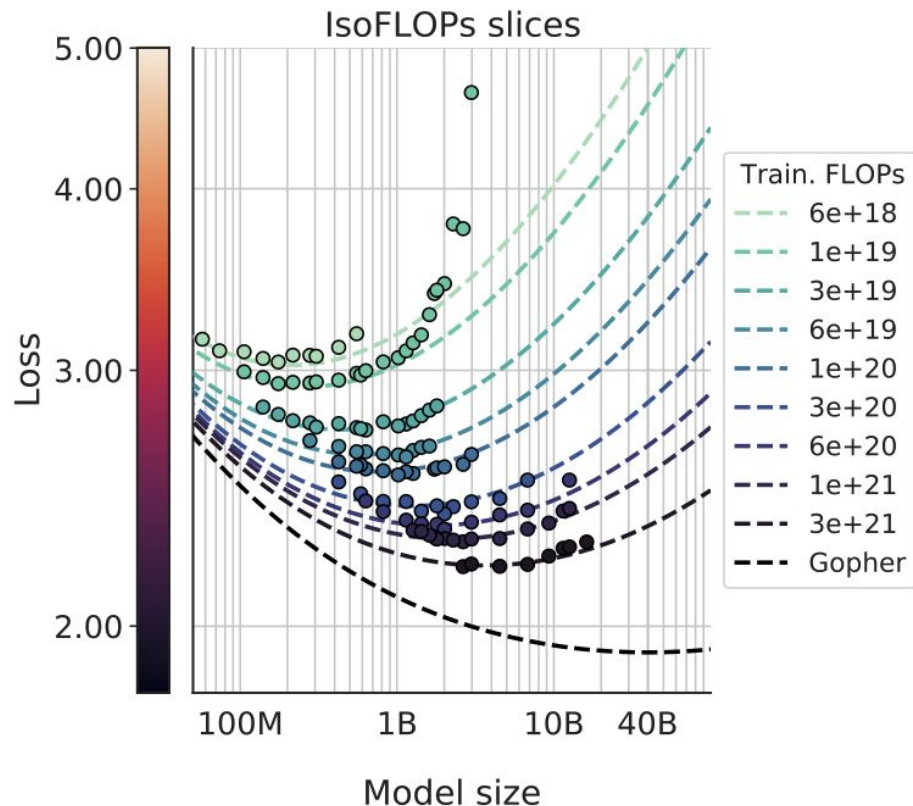
$$\hat{L}(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}$$

Fitting loss scaling parametric form

Fit ~200 models

$$\min_{A,B,E,\alpha,\beta} \sum_{\text{Runs } i} \text{Huber}_{\delta} \left(\log \hat{L}(N_i, D_i) - \log L_i \right)$$

Find $\alpha \approx \beta \approx 0.32$



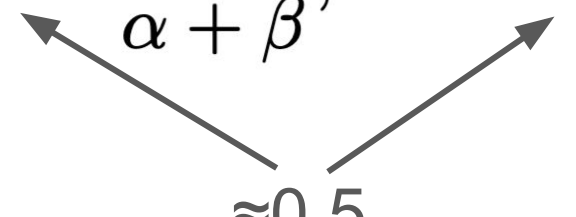
Efficient frontier

$$N_{\text{opt}}(C) = ? \quad \text{and} \quad D_{\text{opt}}(C) = ?$$

Efficient frontier

$$N_{\text{opt}}(C) = G \left(\frac{C}{6} \right)^a \quad \text{and} \quad D_{\text{opt}}(C) = G^{-1} \left(\frac{C}{6} \right)^b,$$

where $G = \left(\frac{\alpha A}{\beta B} \right)^{\frac{1}{\alpha + \beta}}$, $a = \frac{\beta}{\alpha + \beta}$, and $b = \frac{\alpha}{\alpha + \beta}$



≈ 0.5

Tokens per parameter

$$\frac{D_{\text{opt}}}{N_{\text{opt}}} = ?$$

Tokens per parameter

$$\frac{D_{\text{opt}}}{N_{\text{opt}}} \approx 20$$

Group discussion questions

If you have a fixed compute budget and lots of data and potential model size, can you now train the best possible model?

When might these scaling laws not hold?

Group discussion some answers

If you have a fixed compute budget and lots of data and potential model size, can you now train the best possible model?

- How should you set hyperparameters? μ P, CompleteP, and learning-rate and batch-size scaling ([Yang et al., 2022](#); [Dey et al., 2025](#); [DeepSeek-AI et al., 2024](#); [Bergsma et al., 2025](#))
- Over-training smaller models ([Sardana et al., 2023](#); [Gadre et al., 2024](#))
- Predicting loss under any schedule ([Hägele et al., 2024](#); [Tissue et al., 2024](#); [Luo et al., 2025](#))

When might these scaling laws not hold?

- Broken laws, inverse scaling, running out of data ([Caballero et al., 2022](#); [McKenzie et al., 2023](#); [Villalobos et al., 2022](#))
- Data limits and repeated epochs ([Muennighoff et al., 2023](#); [Kim et al., 2025](#); [Hernandez et al., 2022](#))
- Emergent capabilities ([Wei et al., 2022](#); [Schaeffer et al., 2023](#); [Du et al., 2024](#))
- From loss to task accuracy ([Schaeffer et al., 2024](#); [Schaeffer et al., 2025](#); [Ruan et al., 2024](#); [Hu et al., 2023](#); [Bhagia et al., 2024](#); [Patel et al., 2026](#); [Truong et al., 2026](#))
- Data quality ([Sorscher et al., 2022](#); [Mayilvahanan et al., 2025](#))
- Other architectures ([Kaplan et al., 2020](#); [Tay et al., 2022](#); [Gu & Dao, 2023](#); [Beck et al., 2025](#); [Chen et al., 2026](#))
- MoE scaling laws ([Clark et al., 2022](#); [Krajewski et al., 2024](#); [Abnar et al., 2025](#))
- Coupling data and model size ([Kaplan et al., 2020](#); [Hoffmann et al., 2022](#); [Videau et al., 2026](#))
- Transfer learning ([Hernandez et al., 2021](#); [Henighan et al., 2020](#))

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2023

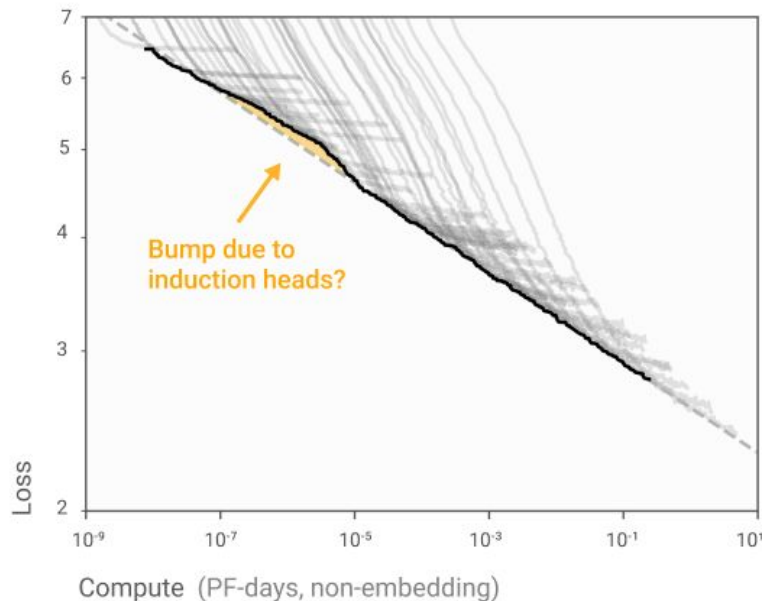
The Quantization Model of Neural Scaling

Eric J. Michaud*, Ziming Liu, Uzay Girit, and Max Tegmark
MIT & IAIFI

Induction head loss drop

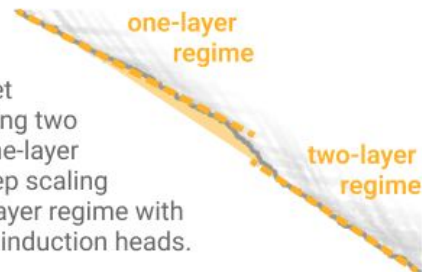
LOSS VS COMPUTE SCALING LAW

Based on Kaplan *et al.*



Kaplan *et al.* observe that “the conspicuous lump at 10^{-5} PF-days marks the transition from 1-layer to 2-layer networks.” We speculate that this is due to the formation of induction heads.

In fact, it's possible that we should interpret the scaling law as having two different regimes. A one-layer regime with a less steep scaling coefficient and a two-layer regime with steeper scaling due to induction heads.



Quanta hypotheses

QH1. Networks must learn a variety of discrete quanta, modules that are either fully learned or not learned, that implementing different algorithm or retrieve different pieces of knowledge.

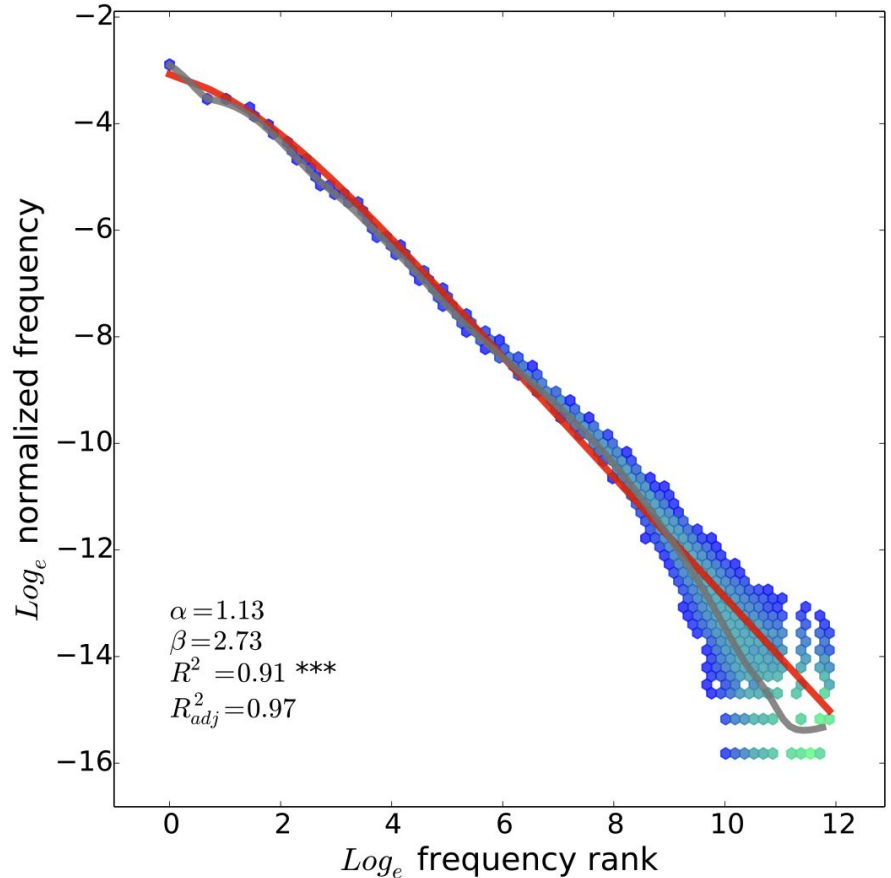
Quanta hypotheses

QH1. Networks must learn a variety of discrete quanta, modules that are either fully learned or not learned, that implementing different algorithm or retrieve different pieces of knowledge.

QH2. Quanta are learned in the order of importance for dropping the loss.

Zipf and real data

Words from the American
National Corpus (ANC)



Quanta hypotheses

QH1. Networks must learn a variety of discrete quanta, modules that are either fully learned or not learned, that implementing different algorithm or retrieve different pieces of knowledge.

QH2. Quanta are learned in the order of importance for dropping the loss.

QH3. Frequencies at which quanta are used for prediction follow a power law.

Quanta hypotheses

QH1. Networks must learn a variety of discrete quanta, modules that are either fully learned or not learned, that implementing different algorithm or retrieve different pieces of knowledge.

QH2. Quanta are learned in the order of importance for dropping the loss.

QH3. Frequencies at which quanta are used for prediction follow a power law.

QH+. Each token needs one quantum, every quantum lowers the loss the same.

Quanta scaling law

$$L(n) = L(0) - \sum_{i=1}^n \Delta \cdot p_i$$

Quanta scaling law

$$L(n) \approx L_{\infty} + Cn^{-\alpha}$$

Data scaling law

$$L(D) \approx ?$$

Data scaling law

$$L(D) \approx L_{\infty} + C_D D^{-\frac{\alpha}{\alpha+1}}$$

Predict scaling exponent from measurable data statistics

Quanta are really hard to measure!

Scaling laws depend on the data distribution.

Approach	Coeff. b where $D_{\text{opt}} \propto C^b$
OpenAI (OpenWebText2)	0.27
Chinchilla (MassiveText)	0.51
Ours (Early Data)	0.550
Ours (Current Data)	0.476
Ours (OpenWebText2)	0.422

Measurable language statistics

Assume models have an effective prediction time horizon $n^*(D)$

You need D large enough to at least detect strongest token-token correlations at temporal separation n .

$$\|C(n)\|_{\text{op}} = O\left(\frac{1}{\sqrt{D}}\right)$$

Loss decomposition

$$L(D) = H_{n^*(D)} + \sum_{n=1}^{n^*(D)} \text{err}_n(D)$$

Power law assumptions

$$H_n - H_\infty = n^{-\gamma}$$

$$\|C(n)\|_{\text{op}} = n^{-\beta}$$

Data limited scaling exponent

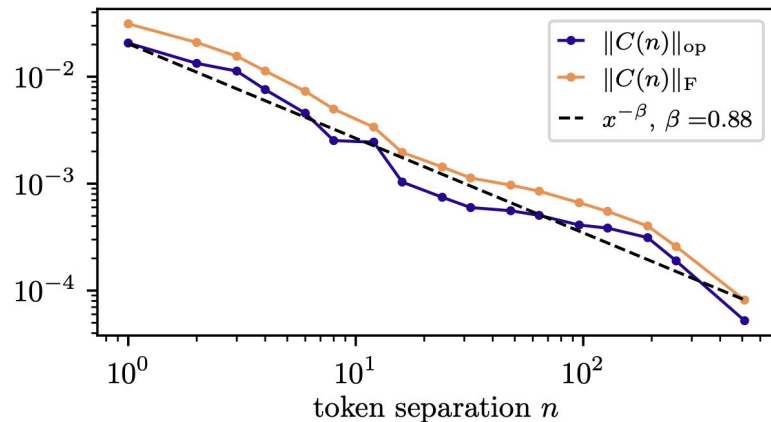
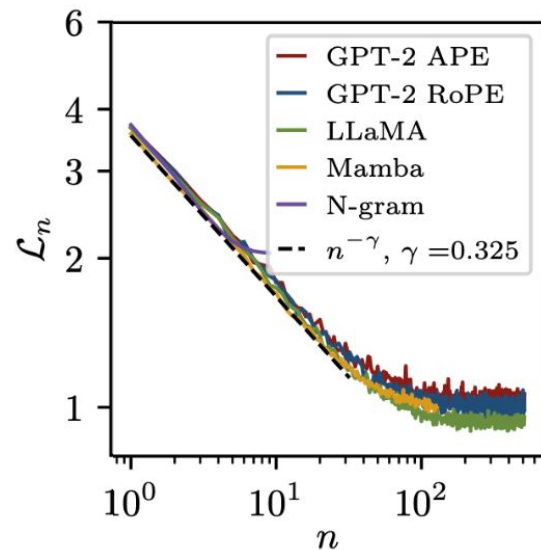
$$L(D) = H_{\infty} + D^{-\frac{\gamma}{2\beta}}$$

Fit γ and β

γ fit by training models

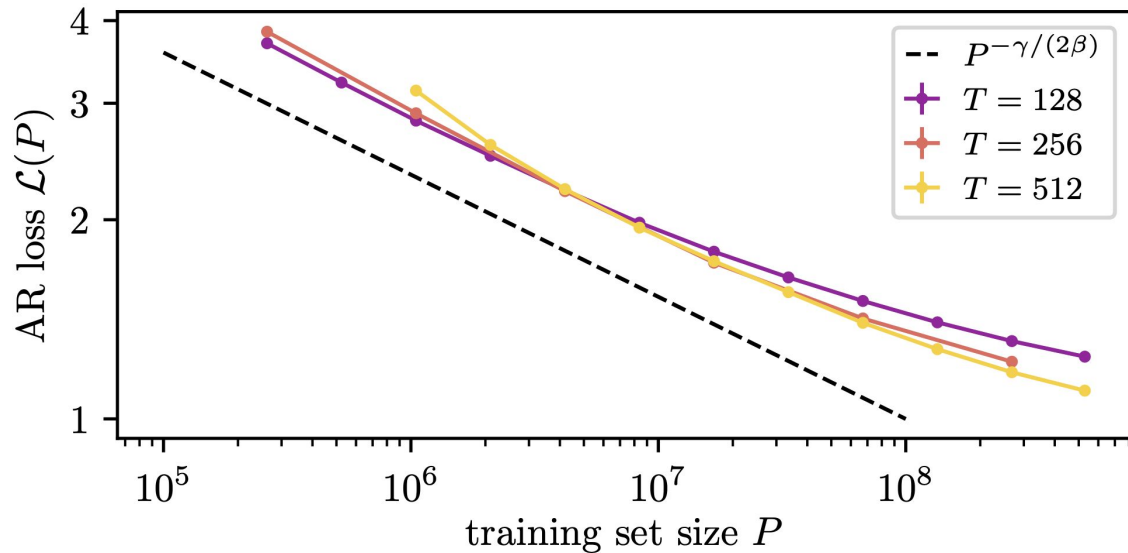
β fit by co-occurrence statistics

Fit on TinyStories dataset $\rightarrow$



How accurate are the exponents?

	TinyStories	WikiText
Prediction	[0.172, 0.198]	[0.116, 0.166]
$m = 5$	[0.1770, 0.1946]	[0.1324, 0.1552]
$m = 6$	[0.1786, 0.1893]	[0.1341, 0.1685]
$m = 7$	[0.1786, 0.1869]	[0.1374, 0.1673]
$m = 8$	[0.1730, 0.1851]	[0.1467, 0.1605]
$m = 9$	[0.1689, 0.1838]	
$m = 10$	[0.1633, 0.1807]	
$m = 11$	[0.1566, 0.1782]	
$m = 12$	[0.1494, 0.1751]	



Group discussion questions

What don't you like about these theories?

What would a better theory of scaling laws look like?

Group discussion some answers

What don't you like about these theories?

- Power law in, power law out ([Piantadosi, 2014](#); [Bordelon et al., 2020](#); [Maloney et al., 2022](#))
- Hard to falsify ([Michaud, 2026](#))
- No joint law
- Mismatched of scaling regimes / bottlenecks

What would a better theory of scaling laws look like?

- Better predict exponents from data statistics at scale ([Learning Mechanics, 2026](#); [Cagnetta et al., 2026](#); [Paquette et al., 2024](#))
- Derive the joint law and the compute-optimal exponent ([Paquette et al., 2024](#))
- More realistic modeling assumptions ([Michaud et al., 2023](#); [Michaud, 2026](#); [Brill, 2024](#))
- Consider superposition ([Liu et al., 2025](#); [Chen et al., 2026](#))

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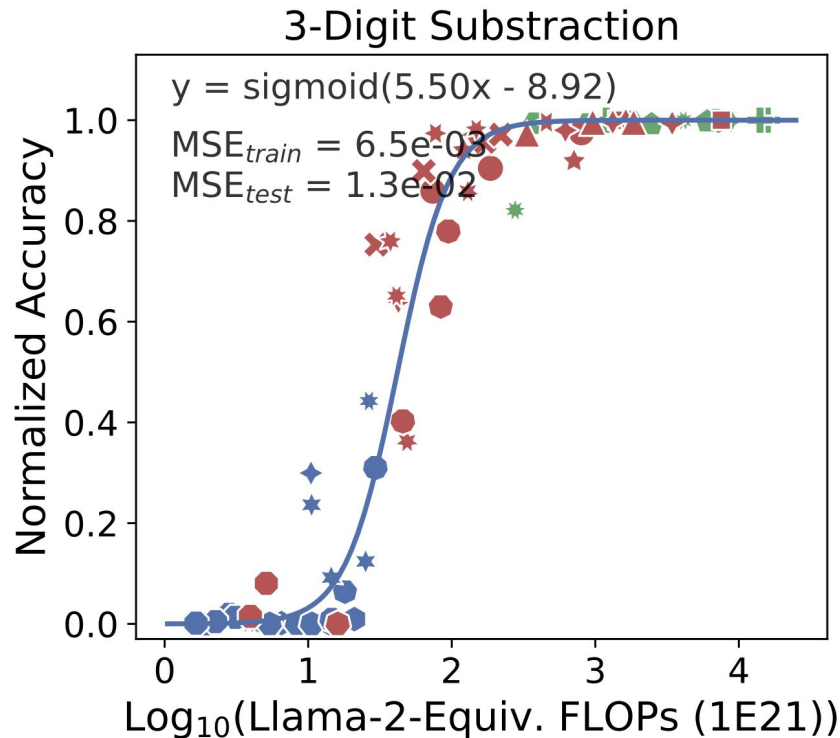
From loss to task performance

Originally we had

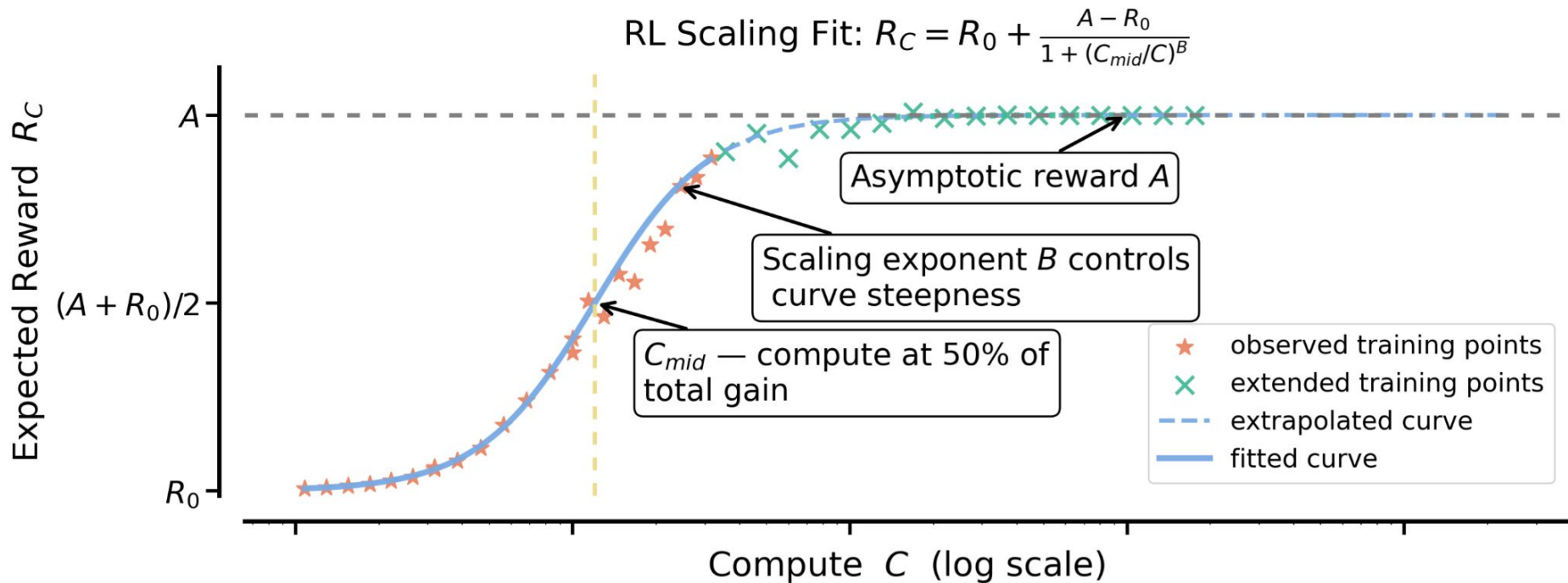
$$\log(L) = \beta \log(D) + B$$

For downstream performance, we have

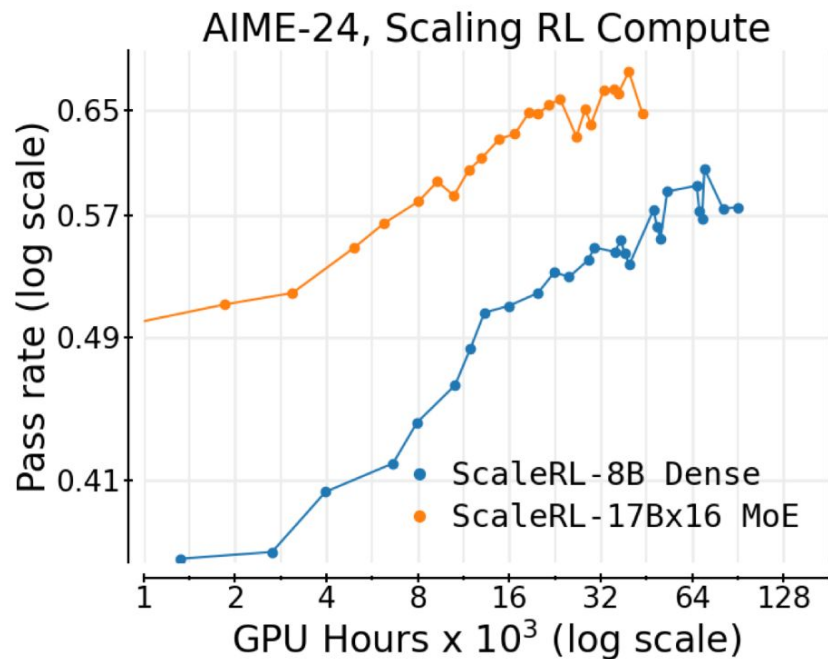
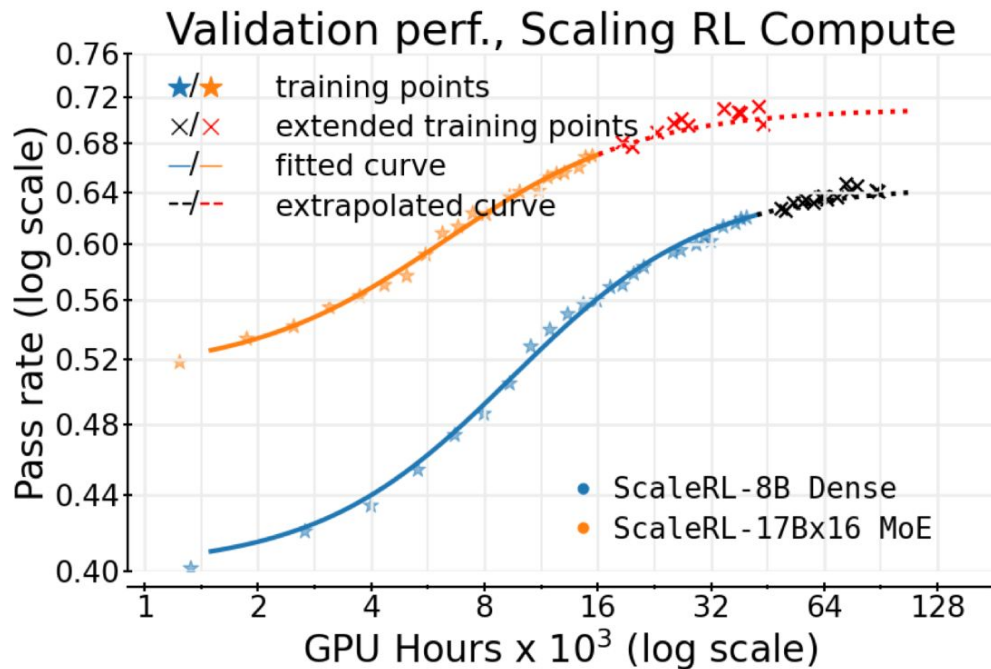
$$\sigma^{-1}(\text{acc}) = \beta \log(D) + B$$



Post-training RL scaling laws



Post-training RL scaling laws fit



Inference time best-of-N single question

Sample independently N times. Success if any of the N responses are correct.

Probability any one question is correct:

$$\text{pass}_i @ N = ?$$

Inference time best-of-N single question

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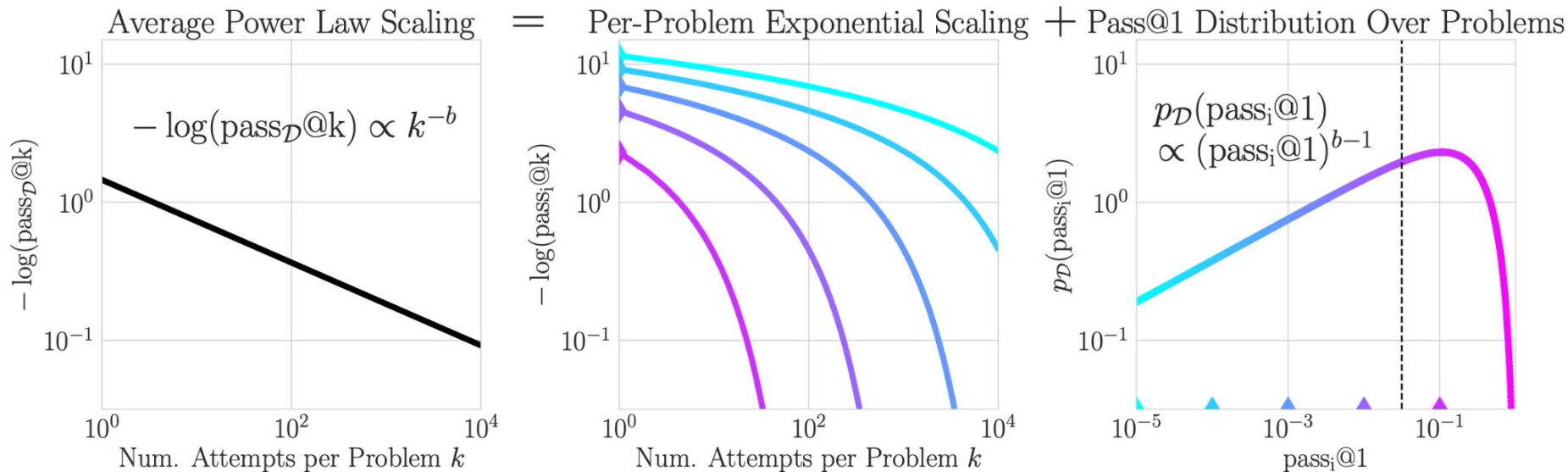
Probability any one question is correct:

$$\text{pass}_i@N = 1 - (1 - \text{pass}_i@1)^N$$
$$-\log(\text{pass}_i@N) \approx (1 - \text{pass}_i@1)^N$$

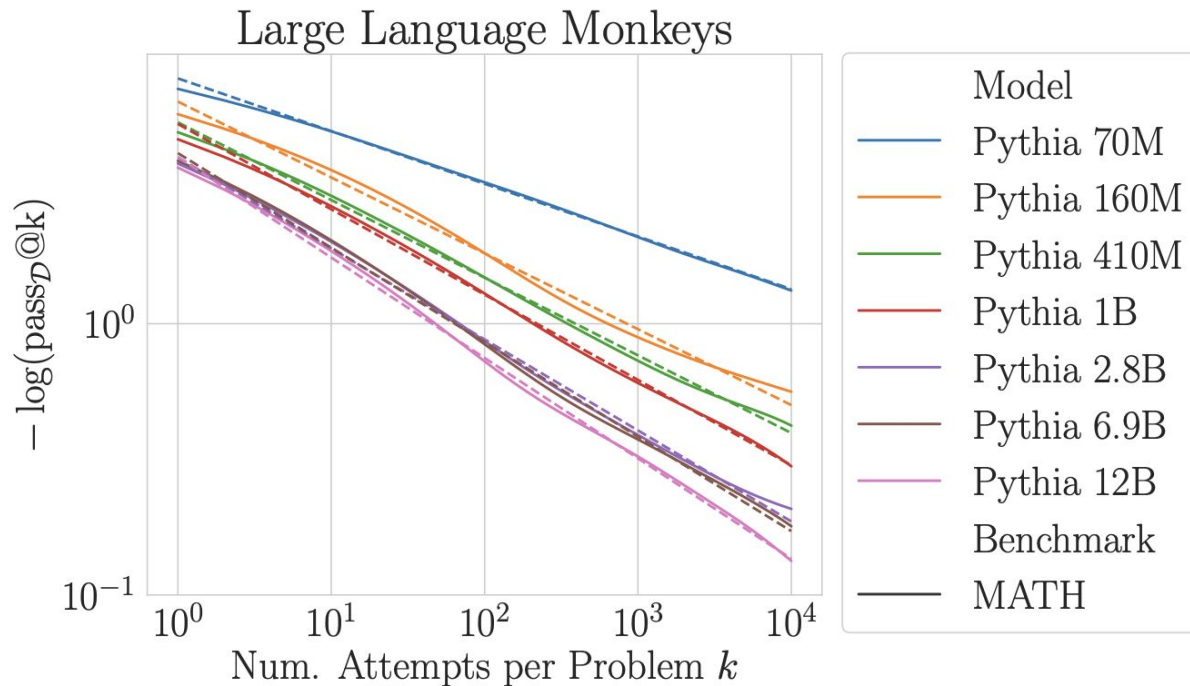
Inference time best-of-N dataset

$$\text{pass}_{\mathcal{D}}@N = \mathbb{E}_{\text{pass}_i@1 \sim \mathcal{D}}[\text{pass}_i@N(\text{pass}_i@1)]$$

Inference time best-of-N dataset



Inference time best-of-N fit



What did we learn?

1. Basics, history, and motivation
2. Empirics for pre-training: Chinchilla empirical scaling laws derivation and fit
3. Theory
 - a. Quanta model: Simple derivations
 - b. Deriving scaling exponent from data: For data-bounded scaling laws
4. Bonus
 - a. Post-training: RL post-training scaling laws
 - b. Inference: Best-of-N scaling laws

Additional references

Scaling laws blog ([Weng, 2026](#))

Lecture from UT Austin ([Golowich, 2026](#))

Thank you